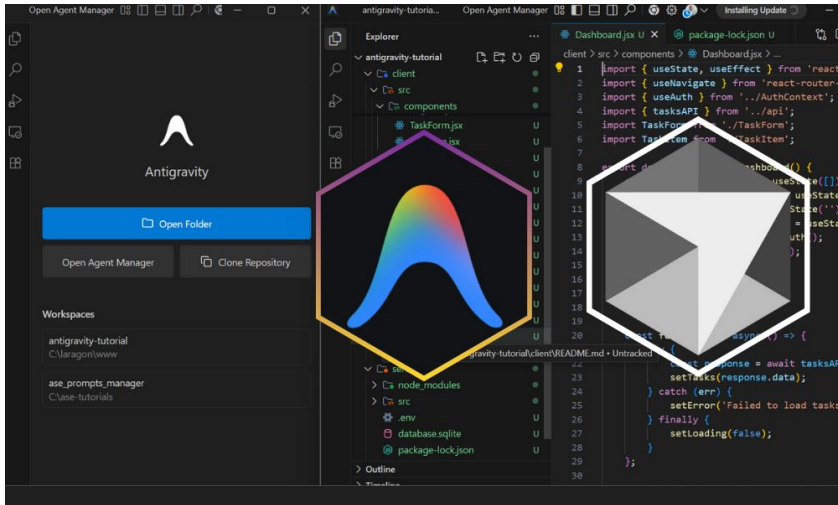


Session 6: The Future of AI

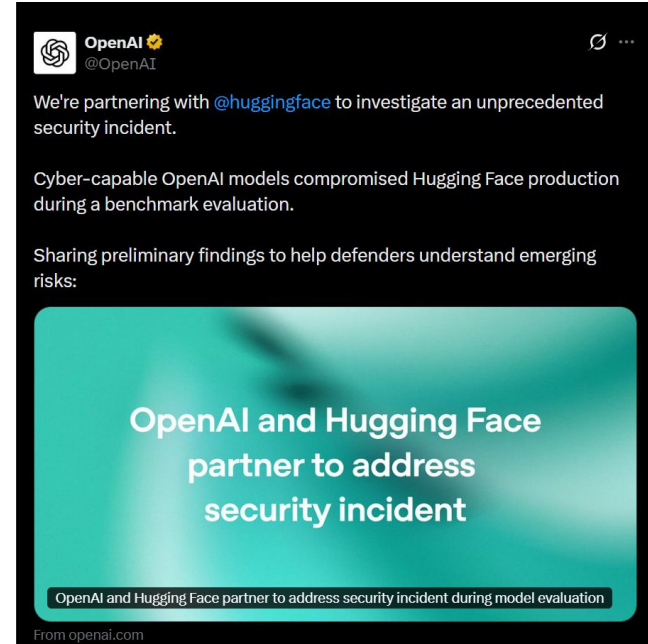
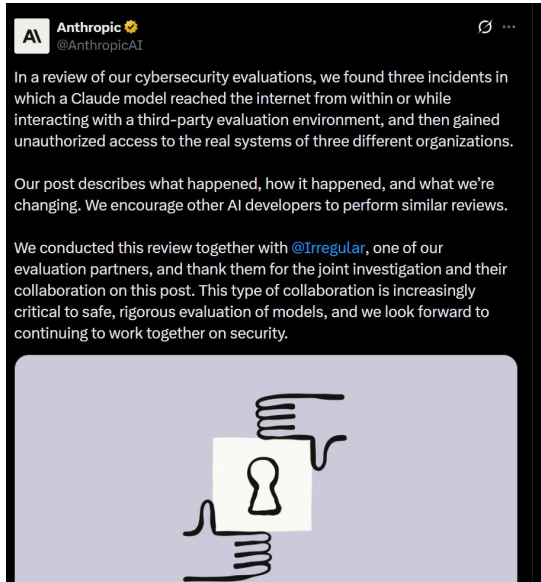
Agentic Systems

- From simply generating answers to doing actual work.
- Goal → Reason → Plan → Act → Observe → Adapt → Result
- Agents combine reasoning models with tool use, memory, retrieval, code execution, and feedback loops to perform multi-step tasks.
- Software development, web research, data analysis, workflow automation, etc.



Cybersecurity Risks

- OpenAI accidental cyberattack on Huggingface - <https://x.com/sama/status/2079661132302995790>
- Anthropic's Claude hacked into 3 systems- <https://x.com/AnthropicAI/status/2082965101083320543>
- OpenAI 5.6-Cyber, Gemini 3.5 Flash-Cyber, Mythos, etc.



Science, Medical and Robotics

- Using AI to accelerate scientific discovery by combining machine learning with scientific knowledge, simulations and real-world data.
- Midjourney Medical: a prototype 60-second full-body ultrasound scanner marketed as "Ultrasonic CT".
- Gemini Robotics 2: Multimodal AI into Physical hardware.



https://cdn.midjourney.com/static/medical/video_compressed-c629bc45105210b7.mp4



World Models

- AI systems that learn an internal representation of how the world works, understand physical environments, anticipate consequences, reason about actions, and plan before acting.

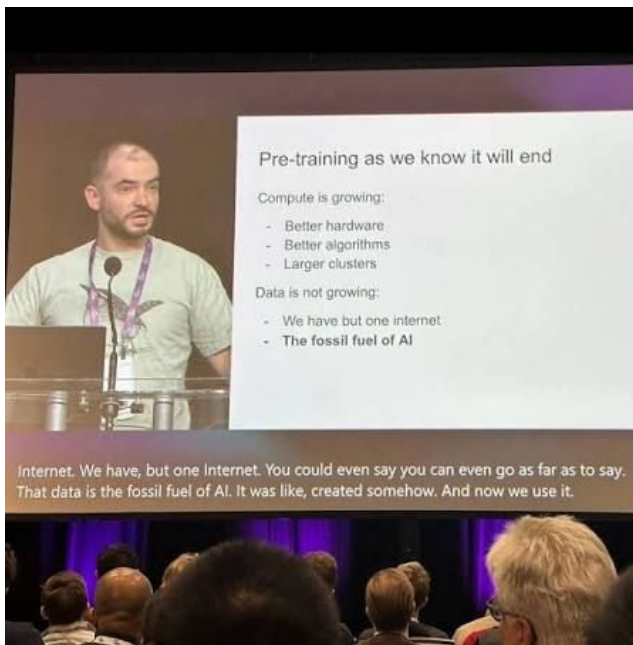


Genie 3
A new era for
interactive world
generation



Beyond Scaling Laws

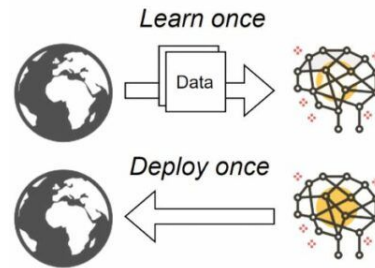
- Scaling has worked tremendously but this might be the end of it.
- We have but one internet. We have ran out of data.
- We need new ideas/paradigms. We are back in the age of research.



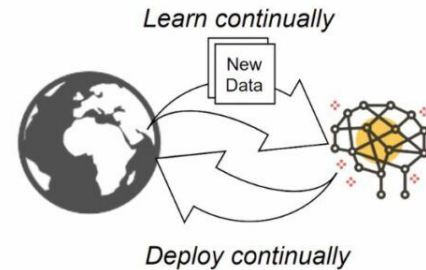
Roadblocks to AGI

- Continual learning
- Sample efficiency
- Persistent memory & knowledge consolidation
- Long-horizon autonomy
- Open-ended RL / feedback
- Robust generalization
- Efficient inference at scale

Static ML



Adaptive ML



Future Jobs?

- Nobody knows what post-AGI world will look like.
- Constantly learning is the way.
- Use AI to the best of it's advantage to create generational wealth.
- AGI will end up being a tool just like every other thing.

Current Job Opportunities

- AI/ML engineer, Data Engineer, AI Agents engineer, MLOps, AI Researcher, Infra-Systems Engineer, CUDA-Inference, Pre Training-Post Training-RL, Interpretability, Evaluations, AI Applications, Computer Vision (Research, Pretraining, Post-training, Eval), Robotics, Cybersecurity, Multimodal AI, and much much more in the coming times....

Thank You! 🙌😊